

CHIRPAN, Bulgaria

WMO: 156350

Lat: 42.200N

Lon: 25.333E

Elev: 178

StdP: 99.20

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.3	-9.3	-14.8	1.1	-11.6	-11.4	1.4	-8.4	12.1	2.9	10.2	4.0	0.6	340	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	34.8	22.7	32.8	22.0	31.3	21.4	24.0	32.5	23.1	30.9	22.2	29.5	2.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.4	16.4	28.3	20.5	15.5	27.1	19.7	14.8	26.1	73.0	32.5	69.4	31.0	66.1	29.4	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.6	7.0	-16.1	38.2	3.9	2.6	-18.9	40.1	-21.2	41.6	-23.4	43.1	-26.2	44.9		
(5)			-16.3	26.0	3.9	1.7	-19.1	27.2	-21.4	28.2	-23.6	29.1	-26.4	30.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	0.4	2.8	7.5	12.1	16.9	21.1	23.6	23.5	19.1	13.1	6.6	1.0	(6)
(7)		DBStd	9.08	4.69	4.95	4.11	3.60	3.16	3.05	2.57	2.75	3.13	3.96	3.99	4.88	(7)
(8)		HDD10.0	1033	297	203	101	21	1	0	0	0	16	116	280	(8)	
(9)		HDD18.3	2673	555	434	337	189	64	10	1	28	166	351	537	(9)	
(10)		CDD10.0	1899	1	2	23	83	216	332	421	419	273	113	14	1	(10)
(11)		CDD18.3	494	0	0	0	1	21	92	163	162	51	5	0	0	(11)
(12)		CDH23.3	5542	0	0	2	23	279	1041	1902	1768	479	48	1	0	(12)
(13)		CDH26.7	2235	0	0	0	2	57	388	855	793	133	6	0	0	(13)
(14)	Wind	WSAvg	2.2	1.8	2.2	2.8	2.9	2.5	2.1	2.2	2.1	2.3	1.8	1.9	1.6	(14)
(15)	Precipitation	PrecAvg	570	44	36	42	43	62	50	54	37	45	41	46	68	(15)
(16)		PrecMax	759	134	76	139	96	145	129	164	87	127	84	137	167	(16)
(17)		PrecMin	389	2	2	3	11	16	19	0	0	5	3	16	20	(17)
(18)		PrecStd	96	31	25	32	23	34	31	44	25	35	25	30	40	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.5	18.2	22.7	26.2	30.3	34.9	38.3	37.1	32.5	27.8	20.1	14.7	(19)
(20)			MCWB	9.1	11.0	13.8	17.1	20.8	22.0	24.3	22.7	21.3	19.8	14.3	11.3	(20)
(21)		2%	DB	10.9	15.3	19.7	23.1	28.3	32.4	35.1	34.8	30.0	24.6	17.4	11.3	(21)
(22)			MCWB	7.5	9.8	12.5	15.3	19.5	21.8	22.8	22.8	20.0	18.0	13.1	10.1	(22)
(23)		5%	DB	8.4	12.7	17.5	21.2	26.4	30.6	32.9	32.8	28.1	22.2	15.0	9.5	(23)
(24)			MCWB	6.4	8.8	11.6	14.4	18.5	21.2	22.0	22.1	19.0	16.3	12.0	8.3	(24)
(25)		10%	DB	6.7	10.1	15.1	19.3	24.5	28.9	31.2	31.1	26.1	20.1	13.2	7.5	(25)
(26)			MCWB	5.3	7.4	10.3	13.5	17.8	20.4	21.5	21.3	18.5	15.4	10.9	6.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	11.7	16.4	18.1	21.6	24.5	25.6	25.2	22.8	21.1	16.0	12.2	(27)
(28)			MCDWB	13.3	16.5	19.9	24.8	28.9	31.8	35.0	34.3	30.2	26.0	19.6	13.9	(28)
(29)		2%	WB	8.2	10.2	13.2	16.1	20.3	23.1	24.0	24.0	21.4	18.9	13.8	10.2	(29)
(30)			MCDWB	10.1	14.4	18.2	21.8	26.9	30.4	32.8	32.4	28.0	23.2	16.2	11.2	(30)
(31)		5%	WB	6.7	8.9	11.8	15.0	19.2	22.0	23.1	23.0	20.2	17.4	12.6	8.7	(31)
(32)			MCDWB	8.3	12.3	17.0	19.9	25.0	28.7	31.0	30.9	25.9	21.0	14.8	9.6	(32)
(33)		10%	WB	5.3	7.5	10.5	14.0	18.2	21.1	22.2	22.1	19.2	16.1	11.0	6.5	(33)
(34)			MCDWB	6.7	10.0	14.9	18.4	23.4	27.6	29.6	29.4	24.3	19.2	13.0	7.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	9.8	11.3	11.5	12.5	13.5	14.0	13.7	12.2	10.5	8.4	6.5	(35)
(36)			MCDBR	11.5	15.1	15.8	15.2	16.0	16.3	16.9	16.5	15.4	14.1	11.8	8.9	(36)
(37)			MCWBR	8.2	10.0	9.4	8.6	8.2	6.8	6.7	6.7	7.2	7.7	7.9	6.6	(37)
(38)		5% WB	MCDBR	10.0	13.8	14.8	13.3	14.4	15.0	15.2	15.2	13.2	11.8	9.8	7.6	(38)
(39)			MCWBR	7.4	9.6	9.3	8.1	8.0	6.9	7.0	7.1	6.7	7.0	6.8	5.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.332	0.362	0.407	0.462	0.446	0.450	0.450	0.459	0.433	0.393	0.361	0.337	(40)	
(41)		taud	2.364	2.285	2.189	2.038	2.165	2.191	2.199	2.170	2.215	2.309	2.357	2.359	(41)	
(42)		Ebn at Noon	799	828	830	808	834	829	824	802	795	782	755	755	(42)	
(43)		Edh at Noon	83	105	130	163	147	144	141	141	126	102	84	77	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.62	2.42	3.46	4.60	5.68	6.22	6.49	5.89	4.36	2.88	1.84	1.40	(44)	
(45)		RadStd	0.22	0.30	0.30	0.51	0.43	0.50	0.49	0.36	0.33	0.30	0.19	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	+0.46	N/A	N/A	N/A	+0.60	+0.60	-108	-156	+76	N/A

Nomenclature: See separate page